Venkata Lakshmi Parimala Pasupuleti

AI Engineer · Multi-Step Agentic Workflows · LangGraph · RAG · LLM Evaluation · HITL · Anthropic · AWS · Python

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PROFESSIONAL SUMMARY

AI engineer with 7+ years production experience specializing in **multi-step agentic workflows, RAG over proprietary data, and LLM evaluation pipelines**. At Intuit, integrates **Anthropic Claude** into financial services, builds **LLM-as-judge eval harnesses** with production monitoring, and implements **guardrails, structured outputs, and observability** across LLM calls and tool invocations. Leasel is an 8-agent LangGraph platform that retrieves over proprietary documents, handles **orchestration, tool use, state management**, escalates via **HITL**, and produces **source-linked outputs** — every answer traced to a cited clause — achieving **93.6% precision, 87.9% F1** across 1,240 test cases. AgentProbe is a live LLM eval toolkit with **dynamic model routing, offline evals, CI regression harness**, and LLM-as-judge scoring across 5 providers. 8 years real estate domain depth at Entrata (1M+ property records, lease compliance, income verification, LIHTC/HCV/HUD workflows). Daily **Claude Code** user. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

Agentic AI: Multi-Step Agentic Workflows, Orchestration, Tool Use & Function Calling, State Management, Memory, LangGraph, LangChain, HITL Escalation, Guardrails, Structured Outputs, Graceful Ambiguity Handling, Source-Linked Outputs, Prompt Caching, Context Engineering, Claude Code

Retrieval & Evaluation: RAG Pipelines, Hybrid Search (TF-IDF + Semantic), Embeddings, Vector Search, ChromaDB, Pinecone, LLM-as-Judge, Offline Evals, Human-in-the-Loop Annotation, Production Monitoring, CI Regression Testing, Hallucination Detection

Backend & Cloud: Python (9+ yrs), FastAPI, Flask, Java, AWS (S3, EC2, RDS, Lambda, SQS, API Gateway, IAM), GCP/Vertex AI, Docker, Kubernetes, GitHub Actions CI/CD, OpenTelemetry, Prometheus, Grafana, PostgreSQL, Snowflake, Redis, MongoDB, SQLite

WORK EXPERIENCE

Intuit

Jun 2026 – Present

Software Engineer — Agentic AI & LLM Systems · Financial Analytics

- Integrate **Anthropic Claude** into Python/FastAPI services for financial data summaries and natural-language queries; implement **guardrails** (input validation, sensitive-data handling, graceful fallbacks when model output fails checks) and **trace-level observability** across LLM calls and tool invocations via OpenTelemetry and Prometheus.
- Build and run **LLM-as-judge eval harnesses** (offline evals + golden datasets + CI regression) and **production monitoring** to measure model and prompt quality before every release — catching hallucination and drift; manage data quality for Snowflake and AWS pipeline assets with structured audit trails; use **Claude Code daily** for scaffolding, refactoring, and debugging.

Entrata — Multifamily & Real Estate SaaS

Jan 2017 – Jul 2024

Senior Software Engineer — Real Estate Data Pipelines & Compliance Systems

- Built production backend systems for **1M+ property records** across a 100K+ MAU multi-tenant platform: lease compliance, income verification, LIHTC/HCV/HUD certification workflows, and income fraud detection — 8 years of **real estate domain expertise** directly applicable to CRE data modeling and retrieval.
- Shipped 20+ features end-to-end (Python, Java, PostgreSQL); async processing (RabbitMQ, idempotency, retry logic); **optimized 200+ SQL queries (30% gain)**; Docker microservices; AWS S3 migration with RBAC for PHI — **100% compliance, zero incidents**; **120+ Agile sprints**.

PROJECTS

Leasel — 8-Agent Agentic AI Platform over Proprietary Real Estate Documents

Agentic · RAG · HITL · Source-Linked

Python · LangGraph (8-agent) · FastAPI · Hybrid RAG (TF-IDF + semantic) · Prompt Caching · SQLite — github.com/venkatalakshmiparimala/Lease_complaince_rag

- Designed and built a **multi-step agentic workflow** (triage → retrieval → reasoning → validation → HITL escalation) with **orchestration, tool use, state management, and memory** over 1,240 proprietary lease documents; **RAG retrieval pipeline** with hybrid search + semantic reranking; **source-linked outputs** cited to exact document clauses; **graceful ambiguity handling** — agent asks for clarification rather than acting on unclear intent; **LLM-as-judge CI eval: 93.6% precision, 87.9% F1**.

AgentProbe — LLM Evaluation Toolkit with Model Routing (Open Source)

Offline Evals · Model Routing · CI Harness

Python · FastAPI · LangGraph · ChromaDB · Anthropic · OpenAI · Azure OpenAI · Groq · Ollama — github.com/venkatalakshmiparimala/AgentProbe

- Dynamic **model routing** across 5 LLM providers (Anthropic, OpenAI, Azure, Groq, Ollama); **offline evals and CI regression harness** with golden datasets; **LLM-as-judge** scoring on factual accuracy, hallucination rate, retrieval quality, completeness, and compliance; 3 LangGraph agents; 15-endpoint FastAPI backend. Evaluates open-weight alternatives on cost, latency, and capability.

AI on FHIR — NLP-to-API Agentic Backend with Fine-Grained Auth

Backend AI · Tool Use · Auth

Python · Flask · React · TypeScript · Docker · BERT · spaCy · LLMs · FHIR REST APIs — github.com/venkatalakshmiparimala/AI_on_FHIR

- Translates EN/ES natural-language queries into **real API actions** via LLMs + BERT + embeddings — agentic tool use over live FHIR endpoints; **fine-grained authorization** (HIPAA-style RBAC) so agents only access what they should; Docker containerized; cut manual lookup time ~60%.

EDUCATION

M.S. Data Analytics Engineering — George Mason University — Graduated May 2026

GPA: 3.87 · Fairfax, Virginia · ML pipelines, LLM evaluation, agentic systems, statistical modeling, data governance

B.Tech Computer Science & Engineering — Jawaharlal Nehru Technological University, Anantapur — 2013 – 2017